

MUHAMMAD SHAHEER ZAMAN SHAH

AI/ML & Agentic AI Developer

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PROFESSIONAL SUMMARY

AI/ML and Agentic AI Developer with hands-on expertise architecting production-grade, multi-agent systems using **LangGraph**, **LangChain**, and **Groq LLMs**. Proven track record building autonomous agentic pipelines spanning enterprise RAG knowledge platforms, AI sales automation engines, real-time computer-vision systems, and multi-agent video/content generation — applying hybrid retrieval (vector + sparse + RRF), state-machine orchestration, and human-in-the-loop design patterns. Computer Science student at FAST-NUCES Islamabad with full-stack delivery experience (React, FastAPI, PostgreSQL/MongoDB) and a strong DevOps foundation, consistently shipping end-to-end systems from model training to production deployment.

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph (state machines, interrupts, checkpointers), LangChain, Multi-Agent Architectures, Groq Cloud API (Llama 3.1 / 3.3)
RAG & Retrieval: Hybrid Search (Dense + Sparse), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, ChromaDB, FAISS, pgvector, BM25
AI / ML / CV: TensorFlow/Keras, MediaPipe, scikit-learn, HuggingFace, Pandas, NumPy
Backend: FastAPI, Node.js, Express.js, REST APIs, WebSocket, Server-Sent Events (SSE), SQLAlchemy, Celery
Frontend: React.js, Next.js (App Router), TypeScript, Tailwind CSS, Vite, Axios
Data & DevOps: PostgreSQL, MongoDB, MySQL, SQLite, Redis, Docker, Alembic, Git/GitHub, Postman
Languages: Python, JavaScript (ES6+), TypeScript, C/C++, Java, Assembly

WORK EXPERIENCE

Web Development Intern

Digital Empowerment Network · Islamabad

July 2025 – September 2025

- Developed and maintained responsive web pages using HTML, CSS, JavaScript, and React.js, improving UI consistency across the platform.
- Collaborated with a team of developers to deliver features on schedule, participating in code reviews and Agile sprints.
- Gained hands-on experience with REST API integration and front-end performance optimisation.

Freelance Web Developer

Fiverr · Remote

2023 – Present

- Deliver custom web development projects for international clients — landing pages, portfolios, and small-business websites.
- Manage the full project lifecycle: requirements gathering, design, development, testing, and client handoff.

PROJECTS

D-VOICE: Real-Time Pakistan Sign Language Recognition System (Final Year Project)

Python · TensorFlow · MediaPipe · FastAPI · React/Vite

- Built an end-to-end real-time PSL recognition platform using MediaPipe landmark extraction and a hybrid TCN+Transformer sequence model for temporal sign understanding.
- Engineered a robust data pipeline with stratified train/validation/test splitting, augmentation, and normalisation parity, processing 188-D feature vectors over 60-frame sequences.
- Achieved 98.25% test accuracy, ~98.25% macro-F1, ~94% mean prediction confidence, and 100% top-3 accuracy across a 9-class active run.
- Developed a full-stack product featuring live sign-to-text translation, speech output, and an interactive Learn module powered by real-time inference.

Agentic AI Visual Novel Video Generator

Python · FastAPI · LangGraph · Groq (Llama 3.3 70B) · React/TypeScript · MoviePy · Pollinations.ai · WebSocket

- Architected a fully autonomous end-to-end video generation pipeline that converts natural-language prompts into complete 50-70 second cinematic visual-novel videos, orchestrated via a LangGraph state machine with four specialised agents: Story, Audio, Video, and Edit.
- Engineered a multi-phase Video Agent featuring AI character sprite generation (Pollinations.ai), background removal (rembg), Ken-Burns animated backdrops, amplitude-driven mouth-sync animation, and multi-scene compositing with libx264 output.
- Built a LangGraph-powered Edit Agent with natural-language intent classification (10 intent types, 95%+ accuracy via Groq), multi-level undo/redo stack, and versioned JSON state snapshots for post-generation editing.
- Implemented a gender-aware TTS Audio Agent (pyttsx3/SAPI5) producing millisecond-accurate timing manifests merged with master audio, fully synchronised with final video output.
- Delivered a React/TypeScript frontend with real-time WebSocket progress streaming, live video playback, and a natural-language edit panel; achieved 66/66 passing tests (45 unit + 21 integration) at 100% free API cost.

NexusIQ — Enterprise Knowledge Agent & Multi-Source RAG Platform

Python · FastAPI · LangGraph · PostgreSQL/pgvector · Groq (Llama 3.1/3.3) · Next.js · Redis

- Architected a 10-node agentic LangGraph retrieval pipeline (intent classification, source selection, query rewriting, parallel retrieval, reranking, hallucination checking) unifying search across Confluence, Notion, Slack, Jira, Google Drive, SharePoint, and PDFs.
- Built a hybrid PostgreSQL search engine combining dense vector search (pgvector HNSW indexes) with sparse BM25-approximate search (GIN trigram/full-text), merged via Reciprocal Rank Fusion (k=60) for optimal retrieval precision.
- Designed hierarchical parent/child chunking — 256-token enriched child chunks for search, 1024-token parent chunks injected into generation context — improving answer coherence and traceability.
- Implemented a cross-encoder reranking stage and self-reflective hallucination guard that validates generated claims against retrieved context before streaming responses via SSE, sustaining sub-2-second end-to-end latency.
- Engineered a parallel, hash-deduplicated ingestion pipeline with batch embedding generation and enterprise-grade audit logging, latency monitoring, and semantic cache telemetry (Redis, cosine threshold 0.95).

NexusSDR — Autonomous B2B Sales Development Representative Agent

Python · FastAPI · LangGraph · ChromaDB/pgvector · Groq (Llama 3.1/3.3) · Next.js · SQLAlchemy

- Engineered an autonomous B2B SDR agent using LangGraph, FastAPI, and Next.js to coordinate end-to-end lead discovery, qualification scoring, sales-context retrieval, and personalised email drafting.
- Modelled the outreach pipeline as a deterministic state machine with checkpointed LangGraph interrupts, freezing graph execution for human-in-the-loop review and edits, ensuring 100% brand safety before send.
- Architected a hybrid RAG retrieval engine combining vector similarity search (ChromaDB/pgvector) and full-text search via Reciprocal Rank Fusion to surface relevant case studies and sales playbooks for personalisation.

- Optimised LLM cost and latency with a dual-model routing strategy — fast model (Llama-3.1-8B-Instant) for parsing/scoring, smart model (Llama-3.3-70B-Versatile) for high-quality email generation.
- Implemented Server-Sent Events (SSE) for real-time backend observability, streaming live execution and scraping status to the Next.js dashboard, and structured the codebase as a production-ready monorepo (api/web/worker apps with shared ORM and schema packages).

Autonomous Multi-Agent Research & Coding System

Python · FastAPI · LangChain · FAISS · BM25 · HuggingFace · Streamlit

- Architected a production-capable autonomous pipeline that converts natural-language tasks into verified, runnable Python code via nine single-responsibility agents: Planner, Researcher, Decision, Coder, Executor, Critic, Reflector, Memory, and Writer.
- Implemented a hybrid RAG retrieval engine combining FAISS dense embeddings, BM25 sparse retrieval, Reciprocal Rank Fusion, and cross-encoder reranking for high-precision document grounding.
- Engineered an iterative Critic/Reflector feedback loop that scores output on functional correctness and LLM-based heuristics, auto-triggering re-generation until a quality threshold is met.
- Built a sandboxed Python execution engine (subprocess isolation, configurable timeout) exposed as a FastAPI REST endpoint returning code, metrics, logs, and a structured report.

Agentic Research Engine

Python · LangGraph · LangChain · ChromaDB · Groq (Llama 3.3 70B) · HuggingFace · Streamlit

- Built an agentic long-form research system using a LangGraph state machine with deterministic node ordering and a conditional retry loop for quality-gated retrieval.
- Implemented a multi-stage pipeline spanning query understanding, planning, query expansion/routing, hybrid dense+sparse retrieval with LLM reranking, context compression, and Markdown report generation.
- Used weighted score fusion (0.7 dense / 0.3 sparse BM25-style) with LLM-based reranking, auto-triggering retrieval retries when relevance fell below threshold.
- Delivered a Streamlit UI with real-time step tracking, animated workflow graph visualisation, and a consulting-style sectioned report renderer.

RAG Customer Support Assistant

Python · LangChain · ChromaDB · Groq · HuggingFace · Streamlit

- Designed a modular RAG pipeline covering document ingestion, chunking, embedding, hybrid retrieval with RRF, reranking, context compression, grounded answer generation, and source citation.
- Built query rewriting and LLM/keyword-based classification to route queries to category-specific document subsets, significantly reducing retrieval noise.

Full-Stack Lyrics Finder & Music Critic App

React · Node.js/Express · MongoDB · Spotify API · Web Scraping

- Built a full-stack music discovery platform with Spotify OAuth, audio feature analysis, and AI-generated music critiques.
- Implemented per-user ratings stored in MongoDB with aggregate statistics; secured endpoints with Helmet, CORS, and rate limiting; deployed live with a CI/CD pipeline.

Trip Management System

React.js · FastAPI · MySQL · Weather API · GPS Location API

- Designed a full-stack trip planner integrating real-time weather forecasts and GPS-based location tracking; RESTful backend built with FastAPI on a normalised MySQL schema.
- Reduced average trip-planning time by enabling one-click weather and route lookup within a single dashboard.

Other Projects

C++ · SFML · Python · TensorFlow · C · POSIX Threads · Cisco Packet Tracer

- **Self-Driving TORCS Racing Car:** Trained a neural network on human-play sensor data achieving consistent autonomous lap completion on 3 test tracks.
- **Escape the Maze:** Procedural maze generation with Dijkstra enemy navigation, custom tree inventory structures, and smooth 60fps SFML rendering.
- **Traffic Management System:** Multi-lane intersection simulation using POSIX threads and mutex synchronisation with zero race conditions.
- **Cisco Enterprise Networking:** Multi-topology design with EIGRP/OSPF/RIP redistribution, NAT, VLSM, ACL, and internal Web & Mail server deployment.

EDUCATION

Bachelor of Science in Computer Science

FAST-NUCES, Islamabad

2022 – Present

Intermediate Certificate (HSSC)

FBISE

2020 – 2022

Matriculation (SSC)

BISE, Abbottabad

2018 – 2020

ADDITIONAL

Languages: English (Fluent), Urdu (Native), Pashto, Punjabi

Activities: Event organiser for multiple university societies; participant in inter-university tech competitions.

Interests: Open-source contribution, competitive programming, reading, traveling, sketching.